INTERNSHIP REPORT

RT-MRI-BASED ORAL CAVITY AREA ANALYSIS AND MAXIMUM ACCELERATION DETECTION

Submitted in partial fulfillment of the requirements for the degree of

MASTER OF TECHNOLOGY (M.Tech.)
COMPUTER ENGINEERING

Submitted by

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Internship Duration: 5 Sept- 5 Nov

Under the Guidance of

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DECLARATION

I, Hitesh Choukiker, declare that this report is a truthful account of the work and learning documented for the project “RT-MRI-Based Oral Cavity Area Analysis and Maximum Acceleration Detection.” I understand that all statements about methods, datasets, results, and conclusions must be supported by my actual work, laboratory records, code, or cited sources.

I will ensure that third-party text, figures, datasets, and methods are appropriately cited; that confidential or restricted research material is included only with permission; and that no unverified numerical result, signature, approval, or synchronization claim is presented as fact.

Any portions of the report prepared with writing assistance will be reviewed, corrected, and accepted by me in accordance with the policies of my academic institution and host organization.

Student: Hitesh Choukiker

Date: _____

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the faculty members and research personnel who supported my learning during this internship. I am grateful to the SPIRE Lab at the Indian Institute of Science, Bengaluru, for the research environment in which I could study the analysis of real-time magnetic resonance imaging (RT-MRI) data related to speech production.

I also acknowledge the academic support associated with my M.Tech. programme in Computer Engineering at SGSITS, Indore. Guidance from supervisors, discussions with lab members, and exposure to scientific workflows helped me understand the importance of reproducible analysis, careful annotation, and transparent reporting.

The work described in this report involves contour representation, anatomical landmark selection, MATLAB-based processing, geometric area estimation, and the planned analysis of temporal changes. I recognize that reliable scientific conclusions depend on validated data, documented assumptions, and independent review. I thank everyone who contributed to the learning process, while accepting responsibility for the accuracy and integrity of the final report.

The names of individual supervisors and collaborators will be added only after confirming the preferred names, titles, and acknowledgement wording with them.

ABSTRACT

Real-time magnetic resonance imaging (RT-MRI) provides dynamic views of vocal-tract configuration during speech production. Quantitative analysis of the oral cavity from such data requires more than visual inspection: anatomical boundaries must be represented consistently, landmarks must be selected reproducibly, and geometric measurements must be computed with explicit assumptions. This project addresses a MATLAB-based workflow for analysing annotated contours associated with RT-MRI frames, with the broader objective of estimating oral-cavity area over time and identifying the time point of maximum acceleration in the resulting area trajectory.

The project workflow uses contour sets referred to as C1, C2, and C3, together with anatomical landmarks such as upper-lip (UL) and lower-lip (LL) points where applicable. The computational pipeline is organized around data loading, preprocessing, contour validation, endpoint identification, interpolation, geometric construction, and temporal analysis. Particular attention is given to duplicate coordinate values, contour ordering, branch selection, common coordinate ranges, and line–contour intersection failures. These issues matter because a contour may contain multiple points with the same horizontal coordinate, and a valid anatomical boundary cannot always be reconstructed by treating the contour as a single-valued function y(x).

Area estimation is treated as a geometric measurement whose definition must match the annotated region and coordinate calibration. If an area sequence A(t) is available at known frame times, numerical differentiation can estimate its rate of change and acceleration. The largest signed acceleration and the largest absolute acceleration are distinct quantities and should be reported separately when relevant. Mapping a candidate event to a phoneme additionally requires a verified temporal relationship between MRI frames and audio; no such synchronization result is assumed in this draft.

The report documents the motivation, background, proposed processing stages, MATLAB implementation considerations, validation plan, result-reporting framework, limitations, and future work. Quantitative performance claims are intentionally left open until actual experiment outputs and calibration metadata are supplied. The report therefore provides a reproducible foundation for finalizing the research record without inventing experimental findings.

Keywords: RT-MRI, speech production, oral cavity, contour analysis, MATLAB, landmark detection, geometric area, numerical differentiation, acceleration, audio–video alignment.

LIST OF TABLES AND ABBREVIATIONS

Abbreviation	Meaning in this report
RT-MRI / rtMRI	Real-time magnetic resonance imaging
MRI	Magnetic resonance imaging
UL	Upper-lip landmark, as defined by the project annotation protocol
LL	Lower-lip landmark, as defined by the project annotation protocol
C1, C2, C3	Contour labels retained from the project's annotation workflow; anatomical meanings must be defined from the protocol
ROI	Region of interest
MATLAB	Numerical computing and programming environment
A(t)	Estimated area as a function of time
v(t)	First temporal derivative of the area trajectory
a(t)	Second temporal derivative of the area trajectory
Δt	Time interval between samples
RMS	Root mean square
QA	Quality assurance

The abbreviations UL and LL are used as project-specific labels. Their exact geometric interpretation, selection convention, and anatomical location must match the lab's annotation instructions.

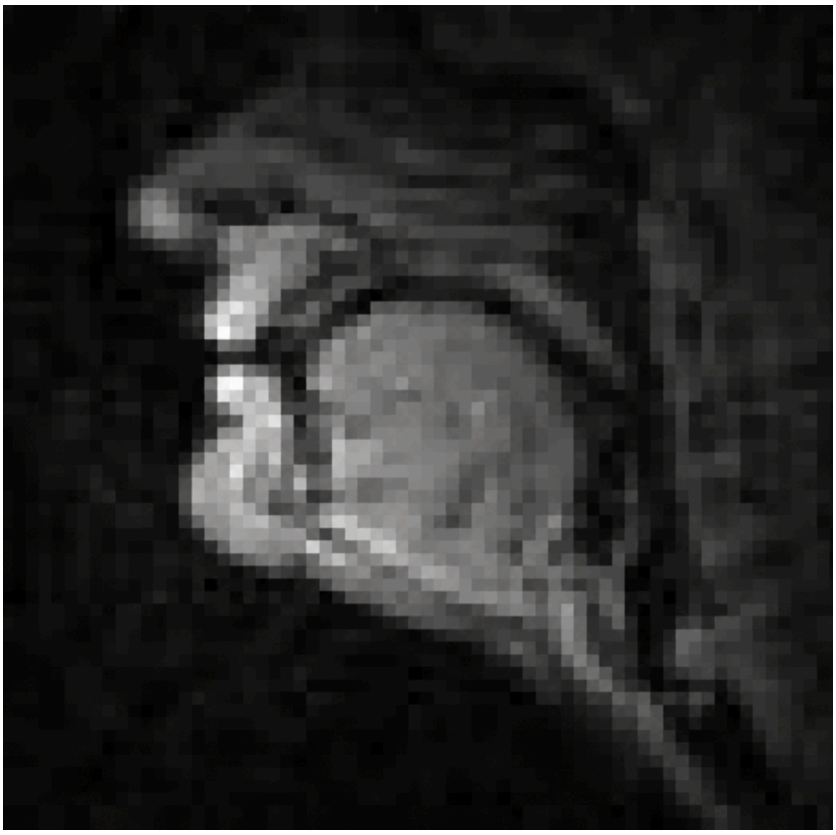
CHAPTER 1 — INTRODUCTION

Speech is produced through coordinated motion of the tongue, lips, jaw, velum, pharynx, and other vocal-tract structures. The configuration of these structures changes rapidly, and the resulting geometry contributes to the acoustic signal. Studying that configuration over time is important in phonetics, speech science, articulatory modelling, and clinical research.

1.1 Research background

Real-time magnetic resonance imaging offers a way to observe vocal-tract motion across a sequence rather than relying only on isolated static postures. Reviews of speech RT-MRI describe its value for examining dynamic articulation and its challenges, including data acquisition, image processing, annotation, and audio–video analysis [1], [2]. The modality can provide useful views of articulators that are difficult to observe simultaneously using some other measurement techniques.

An image sequence, however, is not itself a quantitative description of the oral cavity. A reproducible analysis requires a representation of the relevant air–tissue boundary, a consistent coordinate system, and a defined method for converting annotations into measurements. A contour may be represented as ordered points in image coordinates. Once contours are available, landmarks and geometric relationships can be estimated, but the reliability of the measurement depends on the quality of the annotations and the assumptions made by the algorithm.



RT-MRI FRAME

1.2 Motivation

The immediate motivation for this project is to develop a systematic method for following oral-cavity geometry across time and locating a candidate instant of maximum acceleration in a derived area trajectory. Such a measurement could help characterize rapid changes in articulatory configuration. It should not be interpreted as a complete description of speech motor control: a single scalar area measure discards spatial information and may not uniquely identify a phonetic event.

The project also highlights a practical engineering problem. Contour data are not always clean mathematical curves. Repeated x-coordinates, local loops, uneven point spacing, and multiple branches can invalidate simple interpolation. A robust pipeline must explicitly detect these cases and either handle them according to the annotation convention or flag them for review.

1.3 Project overview

The work is organized as a staged computational pipeline. First, a frame and its associated annotation data are loaded. Second, the contours are checked and preprocessed without silently changing their geometry. Third, reference landmarks and endpoint candidates are identified using a documented geometric rule. Fourth, a region is constructed from the selected boundaries, and an area estimate is calculated if the polygon definition and units are valid. Finally, an area sequence may be differentiated in time to estimate velocity and acceleration.

The available project context includes MATLAB processing of a file named M2_358_temp.mat and a report of 72 available frames for one run. A frame-level debug log records 63 C1 points, 85 original C2 points with one duplicate removed during preprocessing, and 46 original C3 points with one duplicate removed. These are workflow-specific diagnostics, not a complete dataset summary or an accuracy result. They should be retained only if the corresponding run log can be verified and is appropriate for the final report.

1.4 Scope and boundaries

The scope is contour-based analysis of annotated RT-MRI frames, with particular attention to endpoint selection and temporal geometric measurements. It does not assume that the project has achieved fully automatic segmentation, clinical validation, or reliable phoneme identification. Any such claims require separate evidence.

Image-derived area in a mid-sagittal frame is not automatically equivalent to a three-dimensional cross-sectional area or a full vocal-tract area function. Published work on vocal-tract area functions may use 3-D geometry or transformations from sagittal measurements under specific assumptions [3]. This report therefore distinguishes a two-dimensional annotated-region area from a physical three-dimensional area function.

1.5 Report organization

Chapter 2 summarizes the internship and research environment. Chapter 3 formalizes the problem and objectives. Chapter 4 reviews relevant background. Chapter 5 describes data and preprocessing. Chapter 6 sets out the proposed algorithm. Chapter 7 documents MATLAB implementation practices. Chapter 8 defines validation and experimental design. Chapter 9 provides a structured results framework. Chapter 10 addresses temporal and phoneme alignment. Chapter 11 summarizes conclusions and future work. Appendices provide reproducibility and submission checklists.

CHAPTER 2 — INTERNSHIP AND RESEARCH ENVIRONMENT

This chapter places the technical work within the internship context. All organization-specific statements should be checked against official lab information and the internship records before submission.

2.1 Host organization and laboratory

The internship is associated with SPIRE Lab at the Indian Institute of Science (IISc), Bengaluru. The project context identifies research activity involving signal processing, representation, and interpretation of multimodal data. The precise official expansion of the laboratory name, reporting structure, and any permitted institutional description should be copied from authorized host materials rather than inferred.

The research setting is interdisciplinary: image sequences, contour annotations, numerical geometry, temporal signal analysis, and potentially audio information must be treated as connected but distinct data sources. Such projects require coordination between domain assumptions and software implementation. A computational result is only as meaningful as the anatomical definition and data provenance behind it.

2.2 Internship objectives

The internship objective relevant to this report is to investigate a reproducible pipeline for estimating oral-cavity geometry from annotated RT-MRI frames and studying its temporal change. The objectives include understanding the annotation representation, developing or refining MATLAB routines for contour processing, detecting anatomically relevant endpoints, visualizing intermediate decisions, and preparing a validation plan for the resulting area and derivative measurements.

These objectives should be updated to match the final work plan approved by the supervisor. If any task remained exploratory or incomplete, the report should identify it as such rather than describing it as a completed deliverable.

2.3 Work practices and research integrity

A reliable research workflow separates raw inputs, intermediate outputs, parameters, and final figures. Raw annotation data should be preserved. Any preprocessing step that removes duplicate points, reorders a contour, or interpolates an endpoint should be documented and reproducible. Debug logs are useful because they expose the reasoning behind a selection rather than merely reporting the final coordinate.

Research data may be subject to laboratory access rules, participant confidentiality, or publication restrictions. The student should confirm which frames, audio samples, annotations, and screenshots may be included in an academic report. If a figure is not approved for external circulation, it should be replaced with a schematic or omitted.

2.4 Tools and technologies

MATLAB is the principal implementation environment identified in the project context. Its strengths for this workflow include matrix operations, numerical geometry, visualization, and structured data export. The specific MATLAB release, toolboxes, operating system, and dependency versions should be recorded in the final report because they can affect reproducibility.

Other tools may have been used for annotation, image inspection, or audio review. They should be listed only if they were actually used. For each tool, record its purpose, version where practical, input and output formats, and any constraints relevant to reproducing the experiment.

2.5 Expected learning outcomes

The project supports learning in coordinate geometry, contour data structures, interpolation, numerical differentiation, error handling, scientific visualization, and experimental validation. It also illustrates why algorithmic correctness and scientific validity are different: code can execute successfully while computing an area that does not match the intended anatomical region.

An important professional outcome is transparent reporting. The report should distinguish verified observations from proposed methods, failed attempts, hypotheses, and future work. This distinction makes the work more useful to a supervisor or a researcher who may continue the pipeline.

CHAPTER 3 — PROBLEM STATEMENT AND OBJECTIVES

The core problem is to derive a meaningful time series of oral-cavity area from a sequence of RT-MRI frames and then identify a time point associated with maximum acceleration, while preserving traceability from each measurement to the annotated contours and processing decisions.

3.1 Formal problem statement

Given a sequence of image frames indexed by k, corresponding annotation contours, and a known or verified frame-time mapping, define a region Ω_k representing the oral cavity under the project's anatomical convention. Estimate its two-dimensional area A_k and analyze changes in A_k over time. If frame timestamps are available, estimate the first derivative and second derivative of the area trajectory and identify the relevant extremum under a clearly stated definition.

The problem has three linked parts: (i) geometric reconstruction from contours, (ii) temporal measurement from a sequence of frame-level estimates, and (iii) interpretation of an acceleration extremum. Each part introduces uncertainty. An endpoint-selection error can change the reconstructed region; a region-definition error changes area; temporal differentiation can amplify measurement noise; and audio alignment adds a separate synchronization uncertainty.

3.2 Primary objective

Develop and document a systematic, inspectable MATLAB workflow for selecting contour endpoints, constructing the intended oral-cavity region, estimating area over time, and locating a candidate maximum-acceleration event.

3.3 Specific objectives

- Define the coordinate conventions and anatomical meanings of C1, C2, C3, UL, and LL from the annotation protocol.
- Load and validate frame-level annotations while preserving original data.
- Detect duplicate points, invalid values, ordering problems, and ambiguous contour branches.
- Implement endpoint detection using explicit geometric criteria and useful failure messages.
- Use interpolation only when its assumptions are satisfied; otherwise use ordered-curve geometry.
- Construct a region with an unambiguous boundary and calculate its area in the correct coordinate units.
- Record frame timestamps and compute temporal derivatives using an appropriate numerical scheme.
- Validate results using visual inspection, synthetic tests, and independent checks where available.
- Investigate audio–MRI alignment only when reliable timing metadata or a validated synchronization method is available.

3.4 Research questions

The report is guided by several practical questions. Can the relevant contour endpoints be identified consistently across frames? Under what conditions does the chosen geometric construction fail? How sensitive is the estimated area to landmark displacement and contour sampling? Which numerical

differentiation method is appropriate for the available sampling rate and noise level? Finally, what evidence is required before a peak in the acceleration curve can be associated with a particular articulatory or phonetic event?

3.5 Success criteria

Success should be defined through measurable criteria rather than visual plausibility alone. Candidate criteria include: all processed frames have valid contour metadata; endpoint detections are either accepted with diagnostics or explicitly flagged; the area computation passes geometric sanity checks; temporal derivatives use verified timestamps; and reported extrema can be traced to the frame indices and data used to calculate them.

Numerical thresholds must be set from the dataset, annotation variability, and supervisor-approved criteria. This draft does not assign an accuracy threshold or claim that any criterion has already been met.

3.6 Out-of-scope claims

Unless supported by additional experiments, the project should not claim clinical diagnostic value, generalization across speakers or acquisition protocols, automatic phoneme recognition, three-dimensional area estimation from a single 2-D contour, or superiority over published algorithms. The objective is a transparent, validated analysis pipeline for a defined dataset and research question.

CHAPTER 4 — BACKGROUND AND LITERATURE REVIEW

RT-MRI analysis methods vary in the level of representation they produce, from whole-image models to region-based measurements, grid-based area extraction, and explicit contour representations. The choice should be driven by the research question and the available annotations [1].

4.1 RT-MRI in speech research

RT-MRI can provide dynamic information about the vocal tract during running speech. Toutios and Narayanan review advances in acquisition and analysis and discuss the value of the modality for speech science and technology [2]. Nayak and colleagues provide recommendations and technical considerations for RT-MRI speech studies [4]. These sources emphasize that acquisition, reconstruction, audio recording, and analysis choices influence the data that researchers can reliably interpret.

Ramanarayanan and colleagues review a range of analysis techniques and organize them into broad families, including whole-image, ROI-based, grid-based, and contour-based approaches [1]. Contour-based processing is particularly relevant when tissue boundaries or annotated anatomical structures are available and the analysis depends on explicit geometry.

4.2 Contour-based representation

A contour can be represented as a sequence of points p_i = (x_i, y_i). In a digital image, coordinates usually refer to pixels or a calibrated image coordinate system. A contour may be open or closed, and its point order may follow a tracing direction that is not monotonic in x. A robust algorithm must preserve or reconstruct the intended order without sorting away the geometry.

Contour labels such as C1, C2, and C3 are project-specific. Their anatomical meanings should be defined using the lab's annotation guide. The code should not infer the meaning of a contour solely from its identifier. Likewise, UL and LL should be tied to the actual landmark-selection protocol rather than treated as universal labels independent of the dataset.

4.3 Geometric region and area

If the oral-cavity region is defined as a closed planar polygon with vertices (x_i, y_i), the shoelace formula provides a standard area computation: A = 1/2 |Σ(x_i y_{i+1} − x_{i+1} y_i)|, with the final vertex connected to the first. This formula computes the area enclosed by the supplied polygon, not necessarily the anatomically correct oral cavity. Correctness therefore depends on boundary selection, closure, point order, and the meaning of the region.

If coordinates are in pixels, the resulting area is in square pixels. Conversion to square millimetres requires an appropriate image calibration and an understanding of pixel spacing and geometric distortion. If the analysis is performed in a sagittal plane, the result should be described as a two-dimensional projected or annotated area unless a validated transformation to a different physical quantity is used.

4.4 Area functions and 3-D interpretation

Story, Titze, and Hoffman used MRI-derived vocal-tract shapes to study area functions in planes selected relative to the vocal-tract centreline [3]. This work illustrates that an area function along the vocal tract is not identical to a single 2-D region area in a sagittal image. The two quantities answer different questions and require different geometrical assumptions.

Consequently, this project report uses the phrase oral-cavity area only for the explicitly defined region Ω_k. If the eventual scientific objective requires a true vocal-tract area function, additional methodology and validation are necessary. The report should not conflate the two.

4.5 Interpolation and geometric ambiguity

Linear interpolation between two points can estimate a crossing when a signed geometric function changes sign across a segment. If g_j and g_{j+1} are signed distances or line-side values of opposite sign, the crossing fraction can be written as $\eta = g_j / (g_j - g_{j+1})$, and the interpolated point as $p^* = p_j + \eta(p_{j+1} - p_j)$. This formula assumes a valid bracketing segment and a consistent definition of the signed function.

A contour that lies entirely on one side of a test line does not contain a sign-changing crossing under that test. The correct response is not to fabricate a point; it is to verify the reference point, test vector, coordinate system, contour identity, and anatomical assumption. Multiple crossings may occur, so the algorithm needs a selection rule based on anatomy and reference geometry.

4.6 Temporal differentiation

Given area estimates A_k at sample times t_k, a first derivative can be approximated with a finite difference. For equally spaced samples, a central difference estimates velocity near an interior sample as $(A_{k+1} - A_{k-1})/(2\Delta t)$. A second derivative can be approximated as $(A_{k+1} - 2A_k + A_{k-1})/(\Delta t^2)$. These equations are illustrative; irregular timestamps require nonuniform finite differences or a suitable fitted model.

Differentiation amplifies high-frequency measurement noise, and second differentiation is generally more sensitive than first differentiation. Smoothing can help but may also shift or attenuate extrema. Any smoothing method, parameter, boundary treatment, and effect on the location of the peak should be documented. A maximum acceleration should be defined as either the maximum signed value or the maximum magnitude, depending on the scientific question.

4.7 Audio and phoneme alignment

Speech RT-MRI studies may include an audio channel, but synchronization between audio and image frames is a technical issue that requires evidence. Published work explicitly discusses simultaneous acquisition and synchronization challenges [5]. The presence of audio and video files alone does not prove that their clocks, start times, or latencies match.

To associate an acceleration event with a phoneme, a validated temporal mapping must be available, followed by an explicit annotation or alignment method. If that mapping has not been established, the

report should state that phoneme association is a future analysis rather than reporting an unsupported label.

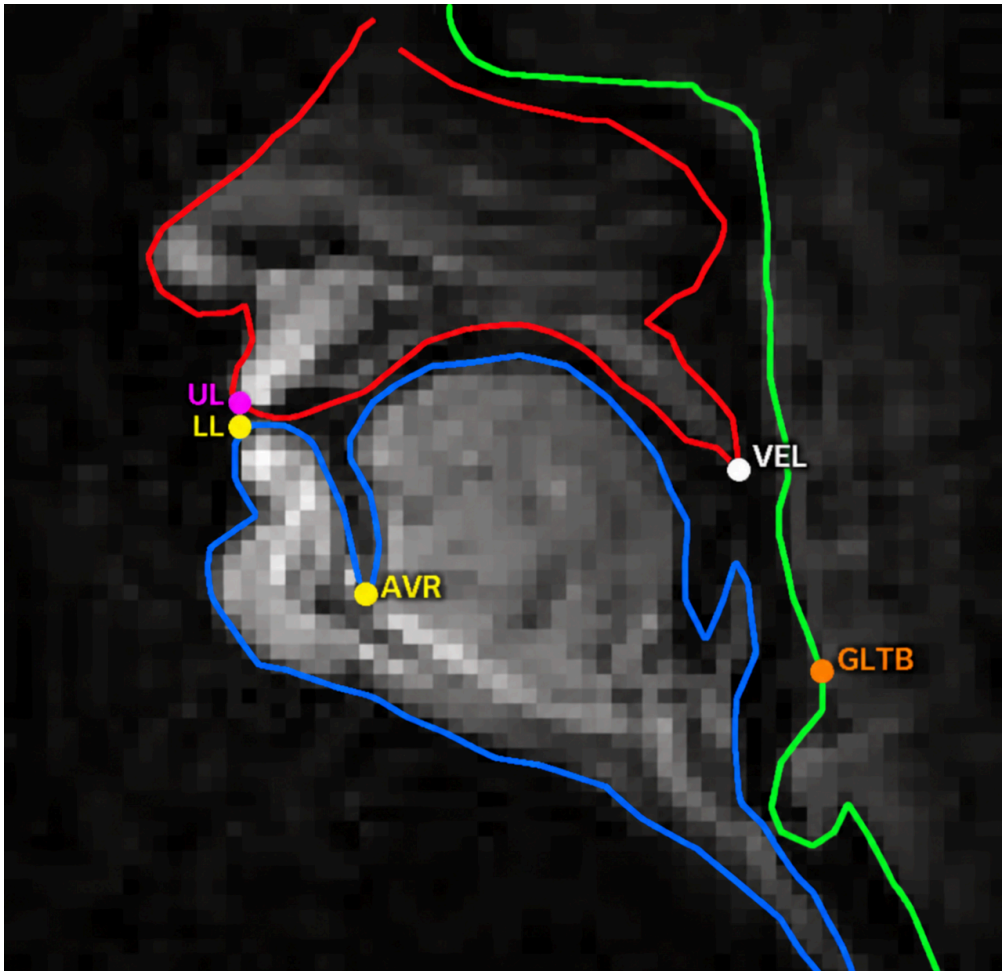
4.8 Research gap and design choice

The practical gap addressed here is not the invention of RT-MRI analysis in general; it is the need for a transparent, dataset-specific pipeline that handles annotated contour geometry, endpoint selection, area estimation, and temporal derivatives with explicit failure states. The chosen design emphasizes traceability and visual review so that each computed measurement can be inspected and reproduced.

The proposed approach should be evaluated against the limits of manual annotations, contour ambiguity, calibration uncertainty, and derivative noise. A successful software pipeline is one that reports uncertainty and failure cases rather than silently producing a number for every frame.

CHAPTER 5 — DATASET AND PREPROCESSING

This chapter describes the data elements required by the analysis. Exact dataset provenance, participant details, acquisition settings, calibration, and file structure must be taken from approved records rather than inferred from filenames.



Annotated RT-MRI Frame

5.1 Data organization

The workflow is based on a sequence of RT-MRI frames and corresponding annotations. The project context includes MATLAB MAT-files and contour arrays. One recorded diagnostic run referenced M2_358_temp.mat and reported 72 available frames. This is a single-run detail and should be included in the final report only if the corresponding log and file are available for verification.

Each frame should have a stable frame index and, where possible, a timestamp. Annotation data should identify which contour belongs to which frame and preserve the original point sequence. The report should document whether annotations are manual, semi-automatic, or generated by another tool, and whether the same annotator or protocol was used across the sequence.

5.2 Annotation labels

The workflow uses C1, C2, and C3 as contour identifiers and UL/LL as landmark labels. These names should be defined in a small annotation schema table that states anatomical meaning, tracing direction, whether the curve is open or closed, and the expected relationship to neighbouring contours. The schema is essential because geometric code can be mathematically correct while interpreting the wrong anatomical boundary.

Anatomical assumptions should be represented in configuration or documentation rather than hidden in variable names. If the meaning of a contour changes between datasets or acquisition conditions, that difference must be handled explicitly.

5.3 Preprocessing principles

Preprocessing should be minimal, reversible where possible, and documented. Non-finite coordinates should be flagged. Exact duplicate consecutive points may be removed if they create zero-length segments, but the raw annotation should remain unchanged and the removal should be logged. Non-consecutive repeated coordinates require more care because they may be legitimate in a loop or contour branch.

Sorting all points by x-coordinate is not a generally valid way to order a contour. It can destroy the tracing path, reorder upper and lower branches, and create false connecting segments. If a downstream operation requires a single-valued y(x) representation, the algorithm must first select the correct branch and address duplicate x-values under a defined rule.

5.4 Duplicate x-values and branch selection

Interpolation using interp1(x, y, x_query) assumes that the x values define a suitable single-valued relationship under the selected method. A contour may contain multiple y-values at the same x because it bends, loops, or contains multiple branches. In that case, using the full point set as if it were a function can produce an error or an anatomically incorrect result.

Safer alternatives include operating on ordered line segments, selecting a branch according to anatomical orientation, intersecting a line with the polyline, or representing the boundary parametrically as p(s) rather than y(x). If duplicate x-values must be consolidated, the rule should be justified and should not merge points from distinct branches simply because their x-values match.

5.5 Coordinate systems and calibration

The report must distinguish image coordinates, display coordinates, and physical coordinates. In many image formats, the origin is at the top-left and y increases downward; in a conventional Cartesian plot, y often increases upward. A transformation may therefore be needed for display or geometry, but it must be applied consistently to all points and vectors.

If pixel spacing is known and isotropic, a pixel-area value can be converted using the squared scale factor. If spacing differs by axis, the area scale factor is the product of the horizontal and vertical spacings. Any crop, resize, rotation, or resampling must be included in the calibration chain. No physical-unit claim should be made without the relevant metadata.

5.6 Data-quality checks

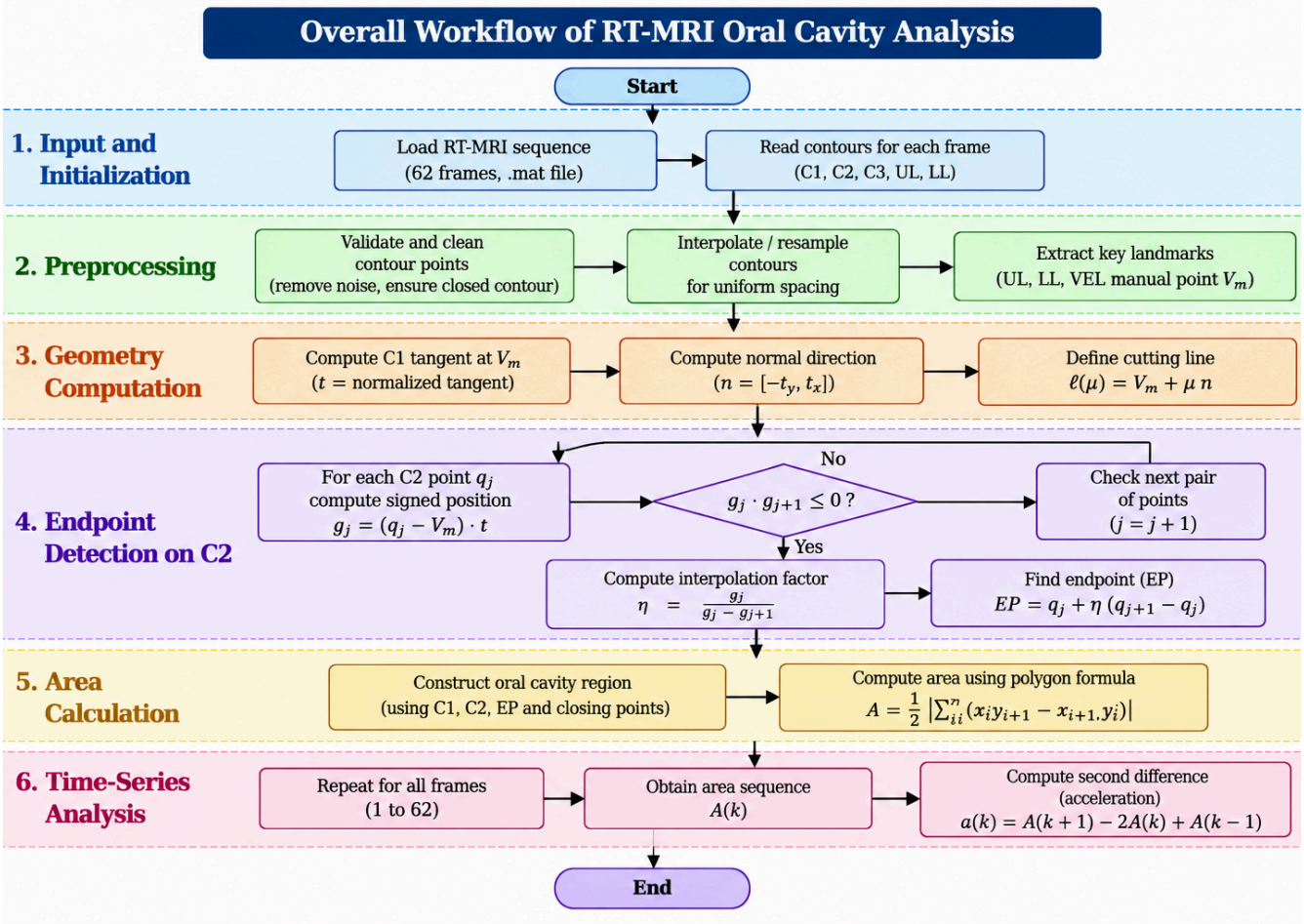
- Confirm that every contour has a valid numeric N-by-2 coordinate array.
- Check for NaN and Inf values and report their locations.
- Count consecutive duplicates and repeated coordinates separately.
- Check for implausibly long jumps between consecutive points.
- Verify contour orientation and expected endpoints against the annotation protocol.
- Record the number of points before and after every preprocessing operation.
- Overlay the raw and processed contours on the source frame for visual inspection.
- Keep a per-frame status field so a failed frame is not silently omitted.

5.7 Privacy and data governance

Only approved frames and annotations should be reproduced in this report. If data contain identifiable information or are subject to a data-use agreement, follow the lab's policy for storage, screenshots, exports, and publication. The report should avoid including sensitive participant details unless they are necessary and authorized.

CHAPTER 6 — PROPOSED METHODOLOGY

The proposed methodology is modular. Each stage produces both an output and diagnostic information, allowing errors to be localized and the processing decisions to be audited.



6.1 Pipeline overview

The high-level pipeline is: load frame and annotation → validate coordinates → preprocess without destroying contour topology → establish reference geometry → detect candidate endpoints → validate anatomical plausibility → construct a closed region → calculate area → attach timestamp → estimate temporal derivatives → identify candidate extrema → perform visual and quantitative validation.

Every stage should define its input, output, assumptions, and failure conditions. For example, endpoint detection should return a structured status such as success, no crossing, multiple ambiguous crossings, invalid contour, or failed anatomical validation. A single error message without frame context makes large-sequence debugging difficult.

6.2 Reference geometry

The project context describes a reference point L computed as the midpoint of manually supplied upper and lower landmarks in one run. In general, if UL and LL are points in the same coordinate system, a midpoint can be defined as L = (UL + LL)/2. This construction is valid only if the two points represent the intended anatomical pair and use consistent coordinates.

A reference direction or test vector may be defined from a manually selected anchor to L, but its orientation and anatomical interpretation must be documented. A direction vector should be normalized if the algorithm uses signed distances with physical distance units; normalization is not required for every sign-only calculation, but consistency is important.

6.3 Signed line-side function

For a line through reference point r with direction vector a, a 2-D signed side function for point p can be defined using the scalar cross product g(p) = a_x(p_y − r_y) − a_y(p_x − r_x). Points on opposite sides of the line have opposite signs, while points on the line have a value near zero subject to numerical tolerance.

A sign-change detector examines consecutive contour vertices and identifies segments where g(p_j) and g(p_{j+1}) bracket zero. It must handle exact or near-zero values, repeated points, and multiple crossings. The test should not assume that every contour crosses every reference line.

6.4 Endpoint interpolation

When g_j and g_{j+1} have opposite signs and the segment is valid, the linear crossing fraction is η = g_j/(g_j − g_{j+1}). The point is p* = p_j + η(p_{j+1} − p_j). A robust implementation verifies that η lies within [0, 1] up to tolerance and that the segment is not degenerate.

If no crossing is found, the code should return a failure state and diagnostic values. The developer should verify the selected contour, line direction, reference point, coordinate convention, and anatomical premise. A fallback rule may be implemented only if it is anatomically justified, documented, and separately validated.

6.5 Multiple-crossing selection

A contour can cross the same line more than once. Choosing the first crossing in array order is not necessarily correct because array order reflects tracing direction rather than anatomical priority. Candidate crossings should be compared against an explicit rule, such as proximity to an expected landmark, branch identity, permitted anatomical range, or a supervisor-approved geometric constraint.

The selection should retain all candidate crossings and the score or rule used to choose one. If two candidates are indistinguishable under the rule, the frame should be marked ambiguous rather than forcing a decision.

6.6 Region construction

The oral-cavity region must be defined before area calculation. A possible construction may combine selected contour segments and interpolated endpoints into a closed polygon, but the exact boundary

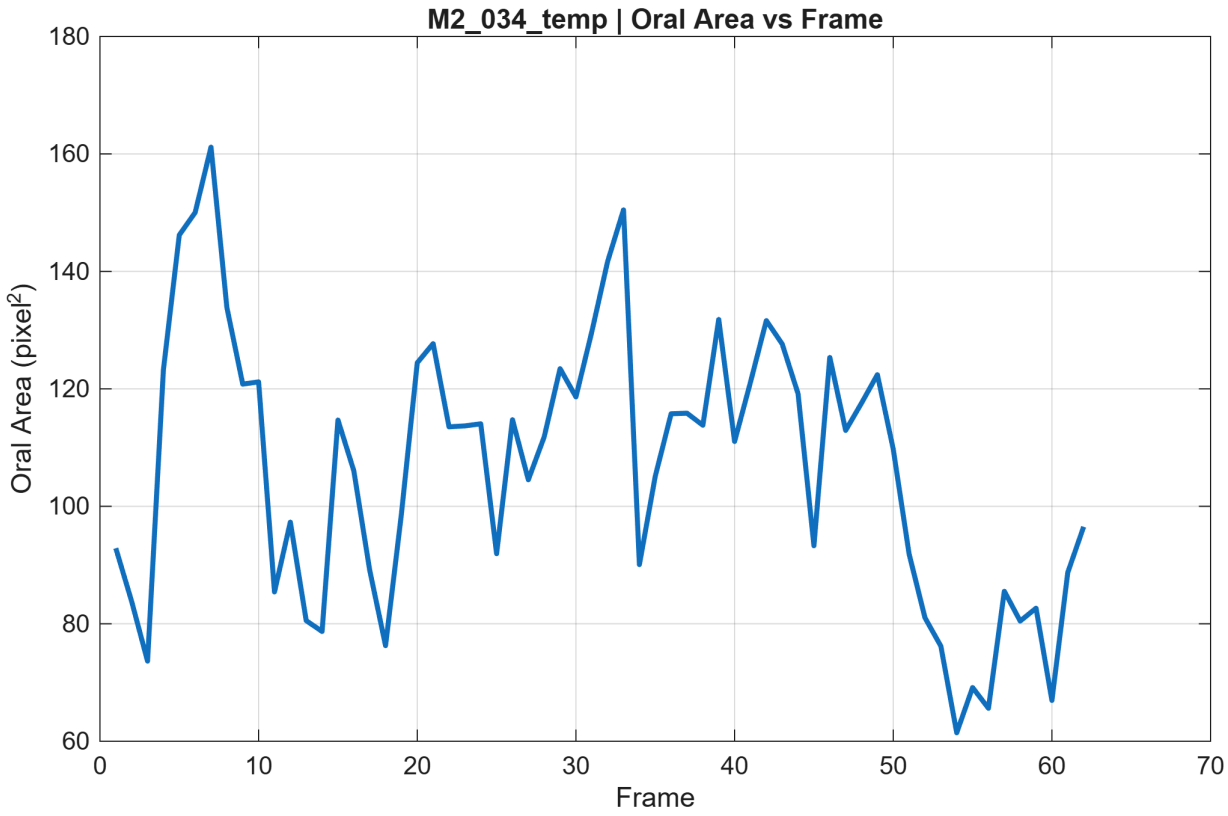
sequence must follow the approved anatomical definition. The algorithm should check that the polygon closes, has sufficient vertices, does not contain unintended self-intersections, and has a plausible area.

The shoelace formula can compute the polygon's signed area. The absolute value gives the magnitude of area but can conceal orientation errors; retaining the signed value during validation is useful. Self-intersecting polygons require special care because the simple formula may not represent the intended anatomical region.

6.7 Area time series

For frame k, store A_k together with frame index, timestamp, region vertices, contour status, units, and confidence or quality flags. Do not fill missing frames with interpolated values without explicitly documenting the imputation. A missing or invalid area should remain marked missing until a justified strategy is selected.

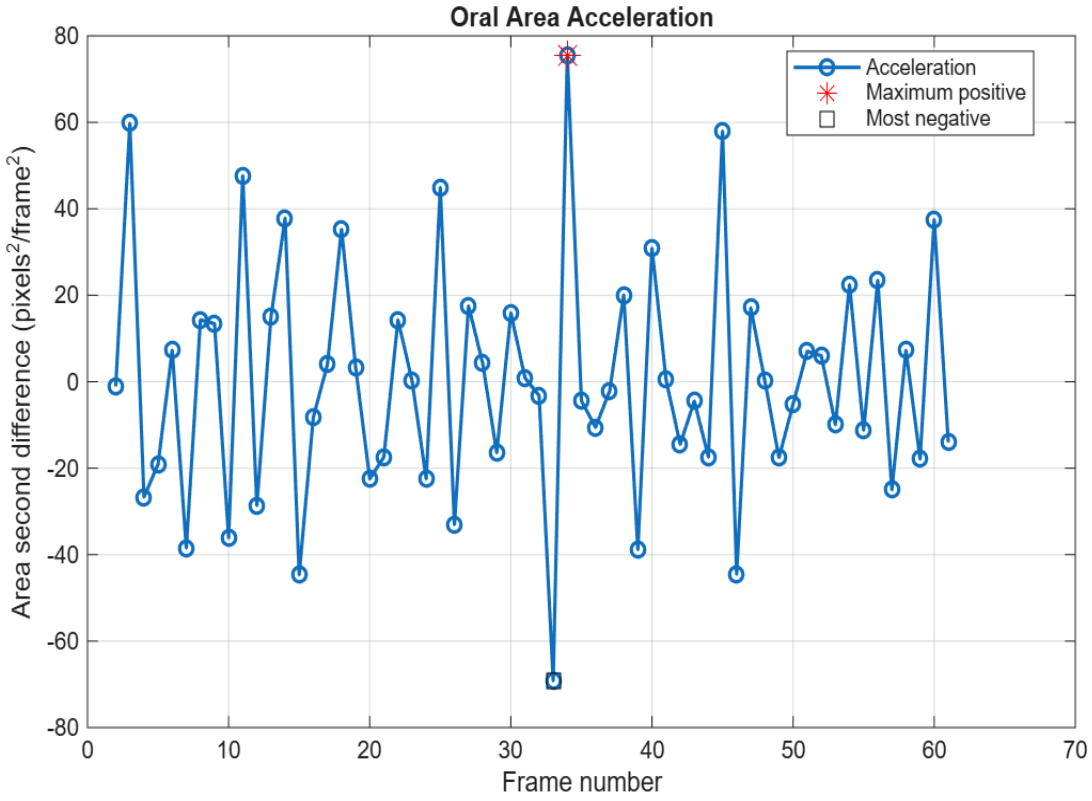
Temporal plots should show the raw estimates and, if smoothing is applied, a clearly distinguished smoothed series. The report should identify the sampling interval and whether it is uniform. If frame timestamps are unavailable, the analysis can be indexed by frame number, but acceleration should not be reported in physical time units.

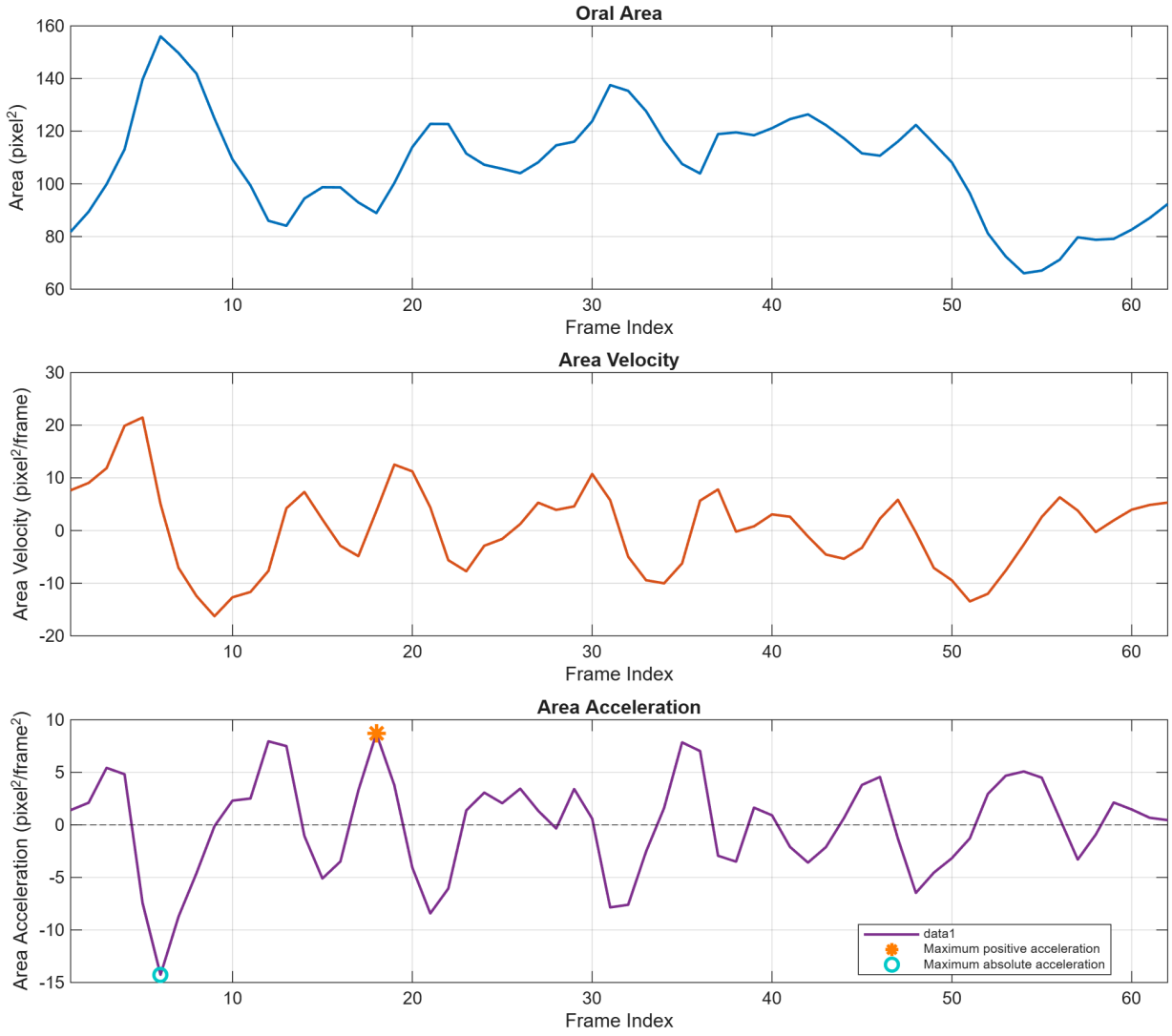


6.8 Velocity and acceleration

For equally spaced timestamps with interval Δt, the central difference estimates the first derivative at an interior frame: $v_k \approx (A_{k+1} - A_{k-1})/(2\Delta t)$. A corresponding second derivative is $a_k \approx (A_{k+1} - 2A_k + A_{k-1})/(\Delta t^2)$. Endpoint estimates require one-sided formulas or should be excluded from extrema selection.

The unit of velocity is area-units per second, and the unit of acceleration is area-units per second squared. If area is in square pixels, these become pixel²/s and pixel²/s². The extremum definition must specify whether it selects max(a_k), max(|a_k|), or another criterion. The peak frame should be cross-checked against neighbouring values and the underlying area trajectory.





6.9 Smoothing and sensitivity

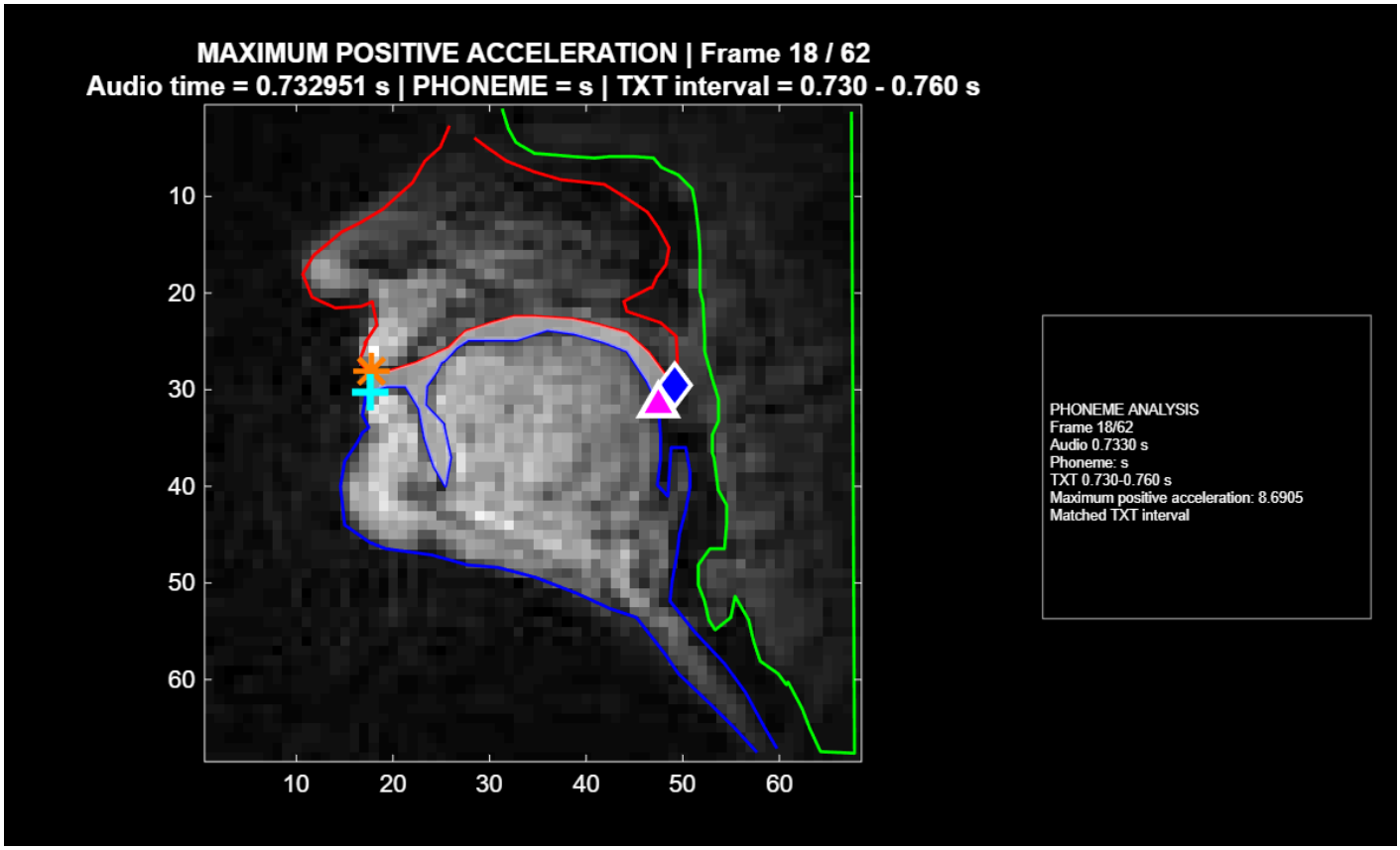
Because second derivatives amplify noise, a smoothing or local-polynomial fitting step may be considered. The report must state the method and parameters and should compare peak location before and after smoothing. A smoothing method should not be chosen solely because it produces a visually appealing peak.

Sensitivity analysis can perturb landmark coordinates within an annotation-informed range, recompute the polygon area, and measure how the selected acceleration extremum changes. This reveals whether the peak is stable or dominated by small contour changes. The perturbation range should be based on known annotation variability or an explicitly justified experimental design.

6.10 Phoneme association

After an acceleration event is detected, phoneme association is a separate analysis stage. It requires a verified map between MRI frame time and audio time, then a phonetic annotation or forced alignment that is suitable for the recording. If synchronization is not validated, the output should be called a candidate event time, not a confirmed phoneme boundary.

Where synchronization metadata are available, record the time offset, frame rate or per-frame timestamps, audio sample rate, and any resampling or trimming operations. A visual or acoustic cross-check may be used as additional evidence, but it should not replace a documented synchronization method.



CHAPTER 7 — MATLAB IMPLEMENTATION

The MATLAB implementation should reflect the methodological separation described in Chapter 6. This chapter records a recommended architecture and the project-specific diagnostic patterns known from the available context. The final implementation description must be checked against the actual source code.

7.1 Modular architecture

A modular implementation is easier to test than one large script. The project context names functions or files such as loadData.m, preprocessData.m, identifyC1.m, identifyC2.m, findSignChangeCrossing.m, findC1Start.m, findC2Start.m, validatePoints.m, saveResults.m, and visualization routines. These names indicate a useful separation of responsibilities, but their final behavior should be verified from the code before being described as implemented functionality.

Each function should have a clear contract. Inputs and outputs should be documented, units should be stated, and errors should include the frame index and contour name. Diagnostic output should be optional or routed through a logger so that large runs do not become difficult to inspect.

7.2 Data loading

The loader should confirm that the MAT-file exists, required variables are present, frame indexing is valid, and each requested contour has the expected dimensions. It should not silently substitute a different frame or reinterpret missing variables. A short summary can report frame count, annotation counts, and the chosen frame index.

The source data should be kept read-only. Any normalized or preprocessed structures should be saved separately with a clear relationship to the raw input and a record of configuration parameters.

7.3 Preprocessing and validation

Preprocessing should return both the processed contour and a diagnostic summary. If one duplicate is removed from a contour, the report should state the rule used and the number of points before and after the operation. A duplicate-removal function should distinguish exact repeated adjacent points from other repeated coordinates.

Validation should check array shape, finite coordinates, degenerate segments, contour length, and endpoint assumptions. The validator should not automatically repair every issue. Some issues require human review because a geometrically unusual feature may be anatomically meaningful.

```
function [isValid, statusTable] = validatePoints(C1, C2, UL, LL, VEL, C2end)
% VALIDATEPOINTS  Verify all four points exist, are finite, and lie
% within/near their source contours' coordinate range (sanity bound).
% Builds the result status table requested.

    fprintf('========== VALIDATION ==========\n');

    names = {'C1 START (UL)'; 'C2 START (LL)'; 'C1 END (VEL)'; 'C2 END'};
    pts   = [UL; LL; VEL; C2end];
    status = cell(4,1);

    for i = 1:4
        p = pts(i,:);
        ok = true;
        reason = '';

        if isempty(p)
            ok = false; reason = 'missing/empty';
        elseif any(~isfinite(p))
            ok = false; reason = 'NaN/Inf coordinate';
        end

        if ok
            status{i} = 'Valid';
        else
            status{i} = sprintf('INVALID (%s)', reason);
        end
        fprintf('%-16s : %s   [X=%.4f, Y=%.4f]\n', names{i}, status{i}, p(1), p(2));
    end
```

Figure 7.1. MATLAB code for contour validation and consecutive duplicate-point removal

7.4 Endpoint detection and the no-crossing condition

The project context includes a run in which C1 start detection found two candidate sign-change crossings and selected one close to the reference. The same run then failed for C2 start detection because all 84 processed C2 points had positive signed values relative to the supplied reference line, with a reported range of approximately 0.092 to 47.759. The code raised a no-sign-change error.

This diagnostic means that the tested C2 polyline did not bracket the reference line under the current signed-side test. It does not, by itself, prove that the contour is invalid or that the line is wrong. The correct next step is to inspect the plotted contour, line, reference point, coordinate conventions, and intended anatomical intersection. If the contour should geometrically cross the line, verify that the correct contour segment and test vector are used. If the contour should not cross it, revise the endpoint definition rather than forcing a crossing.

An implementation should return all candidate crossings and a structured failure record. The failure record should include the reference point, test vector, signed-value range, contour name, frame index, and visual output path.

D:\Downloads\IISC INTERNSHIP\Annotation-gui\Annotation-gui\annotation\result new\localTangent.m

```
function t = localTangent(C1, k)


    n = size(C1,1);

    if n < 2
        error('localTangent:tooFewPoints', ...
            'C1 must contain at least two points.');
    end

    if k < 1 || k > n
        error('localTangent:invalidIndex', ...
            'Tangent index is outside C1.');
    end

    % ---------------------------------------------------------
    % Endpoint: use incoming direction
    % ---------------------------------------------------------

    if k == n

        raw = C1(k,:) - C1(k-1,:);

    % ------------------------|--------------------------------
    % First point: use outgoing direction
    % ---------------------------------------------------------

    elseif k == 1

        raw = C1(2,:) - C1(1,:);

    % ---------------------------------------------------------
    % Interior point: centered tangent
    % ---------------------------------------------------------
```

Figure 7.2. MATLAB implementation of local tangent and normal estimation.

ui.m | report_video_area_details.m | validatePoints.m | visualizeFinalRes

D:\Downloads\IISC INTERNSHIP\Annotation-gui\Annotation-gui\annotation\result new\findC2End.m

```
function [C2end, idxC2end, tauC2end, tangentVec] = ...
    findC2End(C1, C2, VEL, velIdx)



    fprintf('\n');
    fprintf('========== C2 END DETECTION ==========\n');

    n1 = size(C1,1);

    if velIdx < 2 || velIdx > n1
        error('findC2End:invalidVELIndex', ...
            'VEL index must satisfy 2 <= velIdx <= size(C1,1).');
    end

    nBack = 4;

    firstIdx = max(1, velIdx - nBack);

    P0 = C1(firstIdx,:);
    P1 = C1(velIdx,:);

    rawTangent = P1 - P0;

    tangentNorm = norm(rawTangent);

    if tangentNorm < eps
        error('findC2End:zeroTangent', ...
            'Unable to estimate incoming C1 tangent at VEL.');
    end

    tangentVec = rawTangent / tangentNorm;

    fprintf('VEL = (%.4f, %.4f)\n', ...
        VEL(1), VEL(2));
```

Figure 7.3. MATLAB code for C2 line-crossing detection and endpoint interpolation.

7.5 Visualization and debug overlays

Visualization is part of the method, not just a presentation step. For each difficult frame, overlay the original image, raw contours, processed contours, reference line, reference point, candidate intersections, and selected endpoint. Use a consistent legend and distinguish candidates from accepted points. The overlay should not obscure the anatomy.

Save both clean figures for the report and diagnostic figures for development. Any manually adjusted landmark should be recorded as such. A final figure should preserve image scale and avoid misleading transformations or decorative annotations that could be confused with actual anatomical boundaries.

7.6 Result saving

Save a machine-readable output per frame with frame index, timestamp, selected endpoints, area, derivative values where available, units, status, and parameters. Also save a human-readable summary and figures. Use a stable naming convention that connects outputs to the source data without exposing restricted identifiers.

Include software version, run date, configuration, and code revision or hash when possible. This makes it possible to reproduce a result after the script changes.

```
    plotPoint(ax5,C2(idxC2end(2),:),'ms','C2 bracket i+1');
end
% C1 tangent at VEL and its perpendicular cutting line.
t = double(tangentVec(:)).';
t = t ./ max(norm(t),eps);
n = [-t(2),t(1)];
plotLine(ax5,VEL,t,'k-','C1 tangent at VEL');
plotLine(ax5,VEL,n,'c--','Perpendicular C2 cutting line');
plotPoint(ax5,C2end,'m*','C2 END');
title(ax5,'C2 END: tangent + perpendicular cut');

ax6 = subplot(2,3,6,'Parent',fig);
drawMRI(ax6,MRI); hold(ax6,'on');

% Build the oral-area polygon between C1 and C2.
% C1 arc: UL to VEL. C2 arc: LL to C2 END.
% LL is obtained from the updated findC2Start(C2,L,axisA,UL),
% so no-crossing C2 frames use the UL-paired anatomical fallback.
upperArc = getContourArc(C1, idxUL, velIdx, UL, VEL);
lowerArc = getContourArc(C2, idxLL, idxC2end, LL, C2end);
areaPoly = [upperArc; flipud(lowerArc)];

areaValue = NaN;
areaGeometryOK = false;
if size(areaPoly,1) >= 3 && all(isfinite(areaPoly(:)))
    % Calculate oral cavity area in pixel^2.
    areaValue = polyarea(areaPoly(:,1), areaPoly(:,2));
    areaPixels(frameIdx) = areaValue;
    areaGeometryOK = true;

    % Fill is visualization only; C1/C2 are drawn afterward.
    % Therefore the red/green contour overlays do not affect polyarea.
```

Figure 7.4. MATLAB implementation of oral-cavity polygon construction and area estimation.

7.7 Working Steps

1. Load frame sequence and annotation structure.
2. For each frame, validate contours and preserve raw point order.
3. Preprocess according to documented duplicate and invalid-point rules.
4. Build the reference geometry from verified landmarks.
5. Find every valid line–contour crossing for each required contour.
6. Apply the anatomical selection rule; flag ambiguous or missing cases.
7. Construct and validate the region polygon.
8. Compute area and attach units and timestamp.
9. Save diagnostics and per-frame status.
10. After processing the sequence, estimate derivatives using actual timestamps.
11. Identify the chosen acceleration extremum and export a review plot.

CHAPTER 8 — EXPERIMENTAL DESIGN AND VALIDATION

Validation should test the geometry, temporal calculations, and interpretation separately. A pipeline that produces a smooth graph is not necessarily correct; each computational layer needs tests that target its assumptions.

8.1 Validation objectives

The first objective is software validation: does each function implement its stated mathematical rule? The second is annotation validation: do selected points correspond to the intended anatomy? The third is measurement validation: does the constructed region represent the intended area? The fourth is temporal validation: are timestamps and derivative estimates correct? Finally, event interpretation must be assessed independently from the numerical detection.

8.2 Unit tests for line intersections

Create synthetic polylines that cross a known test line once, cross it more than once, touch it at a vertex, lie entirely on one side, contain a duplicate point, and contain a degenerate segment. The expected result for each case should be specified before running the test. The test should check both the coordinate returned and the status code.

Numerical tolerance should be tested around zero. A point very close to the line may be treated as on-line under a tolerance, but the tolerance must not be so large that it changes which branch is selected. Tests should cover positive and negative line orientations to reveal sign-convention errors.

8.3 Unit tests for area

Use simple polygons with known area, including a rectangle, triangle, concave polygon, reversed vertex order, and polygon with repeated adjacent vertices. Verify that the magnitude is invariant to reversing the order for a simple closed polygon, while the signed area changes sign. Test invalid and self-intersecting polygons separately.

Where the region is assembled from multiple contour segments, use synthetic shapes that resemble the expected topology. Verify that segment concatenation does not introduce artificial diagonals or omit part of the boundary.

8.4 Annotation agreement

If a second annotator or expert-reviewed reference is available, compare selected landmarks and region boundaries. Landmark error can be measured as Euclidean distance in pixels or calibrated units. Region agreement can be evaluated with an appropriate overlap measure if both annotations define comparable regions. The report should state the number of frames reviewed and the selection protocol.

If no independent annotation is available, the limitation should be explicit. Visual inspection by the same developer is useful for debugging but does not establish independent accuracy.

8.5 Area plausibility and continuity

Inspect area values for negative values, implausible jumps, sudden discontinuities, or missing intervals. Since a polygon area magnitude is non-negative, negative reported area often indicates a signed-area convention or data-handling error. A large change may reflect genuine articulation, a bad endpoint, a contour-ordering issue, or a frame timing problem; it must be investigated rather than automatically smoothed away.

Continuity is a diagnostic rather than a universal constraint. Real speech movements can be rapid, and an algorithm should not reject all large changes. Thresholds should be set using the sampling rate, expected motion, and validated data.

8.6 Derivative validation

Test derivative code on analytic functions sampled at known times. For example, a quadratic trajectory has a constant second derivative, so a finite-difference implementation can be checked against the known value away from boundaries. Also test irregular sampling intervals if the real data contain them.

Compare candidate derivative estimates under different reasonable window sizes or smoothing parameters. Report whether the peak location is stable. If it changes substantially, the peak should be described as sensitive to processing choices rather than a definitive event.

8.7 Sensitivity and uncertainty

Possible uncertainty sources include manual contour placement, landmark choice, interpolation along a segment, pixel spacing, region closure, frame timing, and derivative smoothing. A sensitivity analysis can perturb one factor at a time or use repeated annotations to estimate how area and peak timing vary.

Uncertainty should be expressed in a form appropriate to the available evidence. If no uncertainty experiment has been run, list the likely sources qualitatively and avoid assigning numerical confidence intervals.

8.8 Acceptance checklist

- Every included frame has a traceable input and processing status.
- All endpoint choices are visually plausible under the annotation convention.
- No-crossing and ambiguous-crossing cases are recorded and reviewed.
- Area calculations pass synthetic geometry tests.
- Physical units are used only when calibration is documented.
- Derivative calculations use verified timestamps and correct units.
- The selected extremum is stable enough for the intended interpretation or its sensitivity is reported.
- Audio–MRI timing is independently verified before phoneme association.

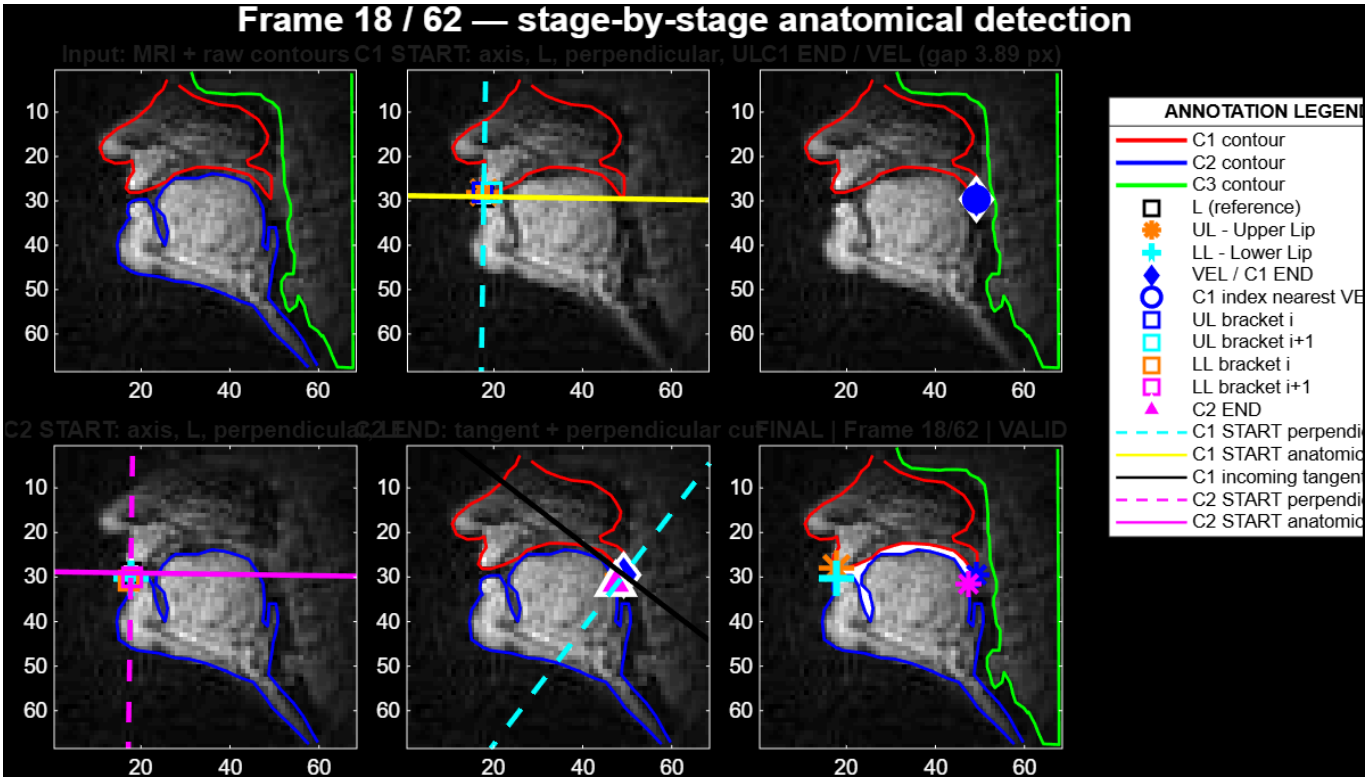
CHAPTER 9 — RESULTS AND DISCUSSION

This chapter combines the project methodology with the final MATLAB output reported by the student for M2_034_temp.mat. The numerical values below are transcribed from the supplied run summary and were not independently re-executed during report preparation. The final submission should attach the saved plots/CSV/MAT outputs and visually validate the contour overlays.

9.1 Processing summary

For the reported final run on M2_034_temp.mat, 62 frames were loaded, 62/62 produced an area value, and zero frames were reported as failed. The saved outputs were reported as M2_034_temp_oral_area_vs_frame.png, M2_034_temp_oral_area_by_frame.csv, and M2_034_temp_oral_area_summary.mat under oral_area_results_indexfix/M2_034_temp. These counts are based on the user-provided MATLAB log and should be checked against the saved CSV/MAT files before formal submission.

The known development log for one frame reports 72 frames available in the MAT-file and the following point counts: C1, 63 points; C2, 85 points before preprocessing and 84 after one duplicate was removed; C3, 46 points before preprocessing and 45 after one duplicate was removed. The final report should label this as a single diagnostic example if the original log is confirmed, not as the result for the full dataset.



9.2 Endpoint detection results

Report endpoint detection performance using a table that includes the frame index, contour, expected endpoint or reference if available, selected coordinate, selection status, and review outcome. If ground-truth endpoints are not available, do not report detection accuracy as a percentage. Instead, report counts of successful, failed, and ambiguous detections, with a clear description of the review procedure.

During an earlier development run, C2 start detection returned 'No sign change found' because all tested C2 points were on the same side of the reference line. This is a geometry/selection diagnostic, not evidence that an endpoint can be safely inferred. The later reported final area run processed all 62 frames, but that does not by itself prove every endpoint is anatomically correct. Preserve the diagnostic and review the corresponding overlay if this case occurs again.

9.3 Area trajectory

The reported area sequence ranged from a minimum of 61.4413 pixel² (frame index not present in the supplied summary) to a maximum of 161.1795 pixel² at frame 7. Selected values were 141.7085 pixel² at frame 32, 150.4925 pixel² at frame 33, 90.0709 pixel² at frame 34, and 105.1177 pixel² at frame 35. The area decreased by 60.4216 pixel² from frame 33 to frame 34. Insert the saved area-versus-frame plot here in the final submitted copy and confirm whether smoothing was used.

An area curve represents the size of the defined 2-D region. It should not automatically be interpreted as airflow, volume, or a complete vocal-tract area function. The discussion should connect geometric changes to anatomy cautiously and should note the limitations of a single scalar measurement.

9.4 Velocity and acceleration results

The reported maximum positive second difference was 75.4683 pixel²/frame² at frame 34; the minimum second difference was −69.2056 pixel²/frame² at frame 33. These are discrete second differences of the area sequence, not physical-time acceleration. The supplied summary does not establish a frame interval or pixel-to-physical-unit calibration. Add the saved derivative plot and inspect the local frames before assigning physiological meaning.

Discuss the influence of sampling interval, smoothing, missing frames, and boundary effects. A narrow peak may reflect a real rapid change, a noisy area estimate, or both. The report should include a local view of the area trajectory around the event and document the neighbouring frames.

Table 9.1 summarises the quantitative values reported in the final MATLAB run. They are presented as reported outputs pending independent verification from the saved result files.

Measure	Reported result	Interpretation / caution
Input MAT file	M2_034_temp.mat	Test sequence
Frames loaded	62	Reported frame count
Valid area frames	62 / 62	0 reported processing failures
Maximum area	161.1795 pixel² at frame 7	Largest reported polygon area
Minimum area	61.4413 pixel²	Frame index unavailable in supplied summary
Area at frame 32	141.7085 pixel²	Selected frame
Area at frame 33	150.4925 pixel²	Before large decrease
Area at frame 34	90.0709 pixel²	Large decrease; visually inspect
Area at frame 35	105.1177 pixel²	Partial rebound
A(34) − A(33)	−60.4216 pixel²	One-frame area difference
Maximum positive acceleration	75.4683 pixel²/frame² at frame 34	Discrete second difference
Minimum acceleration	−69.2056 pixel²/frame² at frame 33	Discrete second difference

9.6 Discussion framework

The final discussion should answer each objective in turn. Did the implementation consistently identify the intended contour endpoints? Which geometric cases required manual review? Did area estimates pass the validation tests? Was the acceleration extremum robust to reasonable analysis choices? Was temporal mapping sufficient to support phoneme association? The answers should be evidence-led and may include incomplete or negative findings.

Unexpected results should be documented rather than hidden. A failed intersection, unstable peak, or unclear anatomical boundary can be a meaningful engineering result because it identifies the limits of the current method and guides the next iteration.

9.7 Threats to validity

Potential threats include annotation bias, limited frame coverage, uncertain pixel calibration, inconsistent coordinate conventions, frame-to-frame tracking errors, region closure assumptions, smoothing-induced peak shifts, and lack of independent ground truth. If only a single speaker or recording is available, generalization is limited.

Results may also depend on how C1/C2/C3 and UL/LL are defined. The report should include the annotation protocol or a concise approved summary so that readers can interpret the geometry correctly.

CHAPTER 10 — TEMPORAL ANALYSIS AND PHONEME ALIGNMENT

Temporal interpretation requires an explicit time base. This chapter separates the computation of an acceleration extremum from the more demanding task of associating that event with an audio segment or phoneme.

10.1 Frame timestamps

Frame indices are not necessarily equivalent to timestamps. If the frame rate is constant and verified, frame time can be derived from the start time and frame interval. If timestamps are stored per frame, those values should be used directly. Dropped frames, variable frame rates, or edited sequences can invalidate a simple index-to-time conversion.

Record the origin of the time base, the frame rate or timestamps, and any trimming or resampling. If only frame indices are available, report the event as a frame index and do not assign a physical time without a verified mapping.

10.2 Numerical derivative choices

Central differences use both preceding and following samples and are generally more accurate for smooth signals at interior points than simple one-sided differences. They cannot be used at the sequence boundaries without a special rule. For nonuniform sampling, derivative formulas must account for the actual time differences.

Alternative methods include local polynomial fits or spline-based derivatives. These methods introduce choices such as window width, polynomial order, or smoothing parameter. The final report should justify the method based on sampling and noise, and compare methods when the location of the peak is a key result.

10.3 Defining the event

Maximum signed acceleration is the largest numerical value of a_k. Maximum absolute acceleration is the sample with the largest magnitude |a_k|, which may correspond to strong negative acceleration. These are not interchangeable. The scientific question should determine which one is relevant, and both may be reported for transparency.

The event should be accompanied by a local plot of area, velocity, and acceleration and a table listing the frame index, timestamp, area, velocity, acceleration, and any quality flags. If the peak occurs near a rejected frame or a sequence boundary, it requires additional caution.

10.4 Audio synchronization requirements

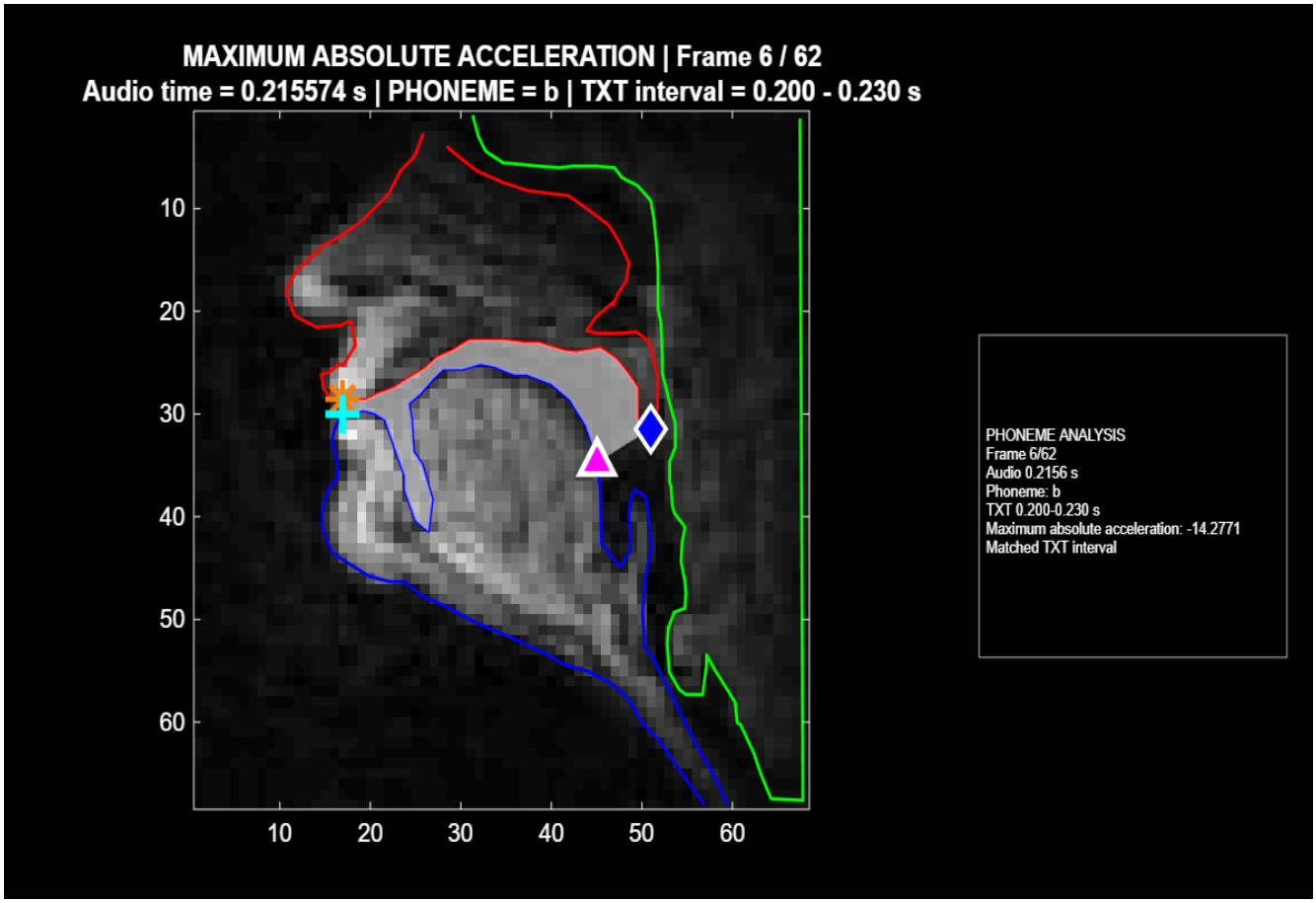
Before phoneme alignment, confirm whether audio and MRI were recorded on a common clock or whether a known offset is available. Record the synchronization method, timing resolution, audio sample rate, MRI frame timing, and any correction applied. Published work identifies synchronization as a distinct technical challenge in real-time speech MRI [5].

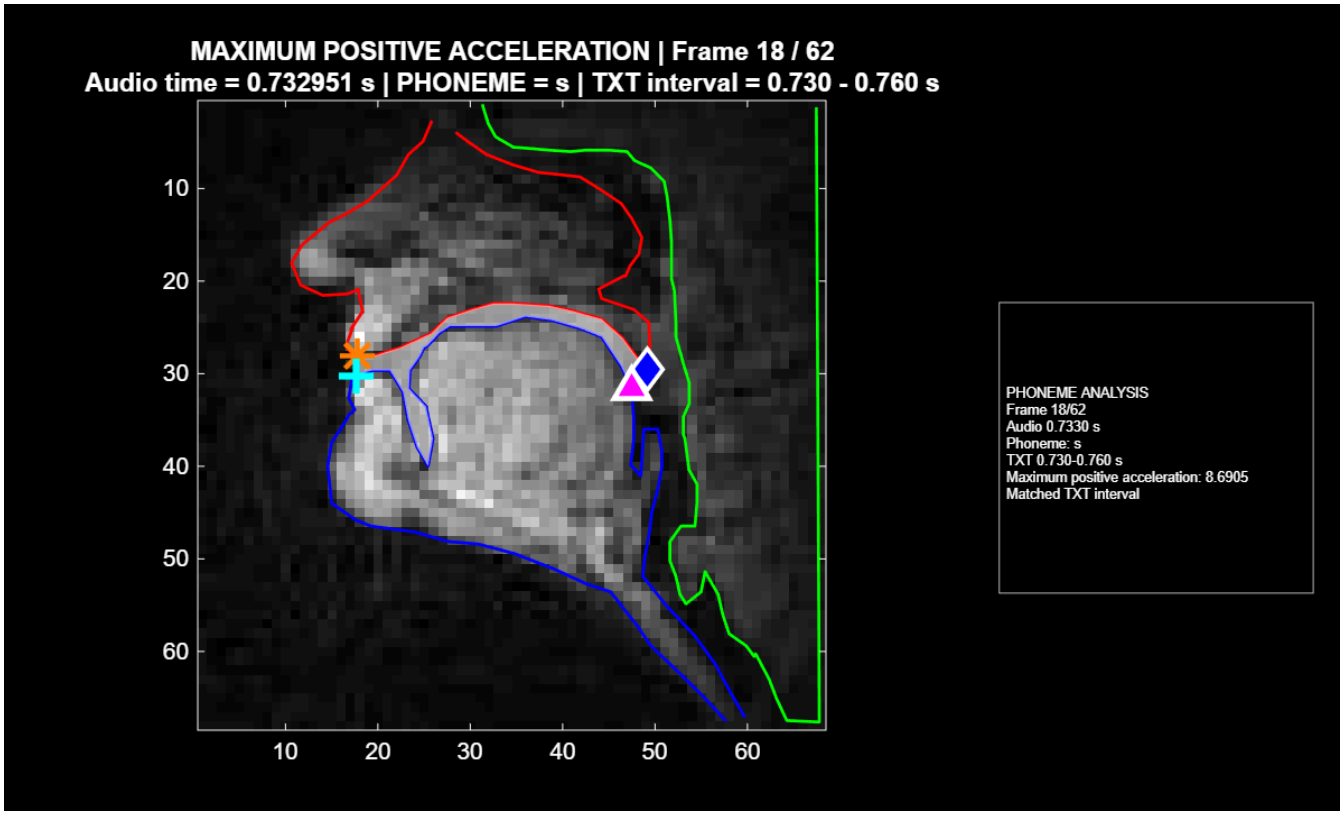
If the recording contains a synchronization pulse, clap, acoustic event, or metadata marker, the method used to locate it should be documented. If no reliable marker or metadata exist, do not infer a precise offset from visual similarity alone.

10.5 Phoneme labelling workflow

Once time alignment is validated, obtain phonetic labels from a trusted manual annotation or an appropriate alignment tool. Record the source of the transcript, the language and phonetic convention, the annotation tool, and whether the labels were manually reviewed. Forced alignment outputs should be treated as estimates and reviewed around the target event.

An acceleration extremum may fall within a phoneme, near a transition, or outside an obvious segment boundary. The interpretation should account for coarticulation and the fact that articulatory motion does not map one-to-one to phoneme labels. The report should avoid claiming that a single peak uniquely identifies a phoneme.





10.6 Reporting alignment uncertainty

Report the time uncertainty of the alignment when it can be estimated. If the event time has uncertainty δt_event and the audio mapping has uncertainty δt_sync, the combined uncertainty depends on their statistical relationship; a simple root-sum-square combination is justified only under suitable independence assumptions. Otherwise, report the components separately.

Where synchronization remains unresolved, the correct conclusion is that phoneme-level interpretation is pending. The geometric analysis can still be reported independently, with the limitation stated clearly.

CHAPTER 11 — CONCLUSION AND FUTURE SCOPE

This project establishes a framework for a traceable MATLAB workflow to analyse annotated RT-MRI contours, estimate a defined two-dimensional oral-cavity region, and examine its temporal change. The final conclusions must be updated after the actual dataset has been processed and the validation plan completed.

11.1 Summary

The report has described the motivation for quantitative analysis of RT-MRI, the role of contour-based representations, the importance of endpoint selection, and the need to distinguish a 2-D annotated region from a 3-D vocal-tract area function. It has proposed a modular processing sequence covering data loading, preprocessing, reference geometry, endpoint detection, region construction, area computation, numerical differentiation, and optional audio alignment.

The project context demonstrates why explicit diagnostics matter. A line–contour crossing detector can find multiple crossings for one contour and no crossing for another. Such outcomes should be interpreted through the geometry and anatomical convention rather than hidden by an arbitrary fallback.

11.2 Conclusions supported at this stage

At this draft stage, the defensible conclusion is methodological: robust area analysis requires reliable annotation semantics, preservation of contour topology, explicit intersection logic, correct coordinate calibration, and validated temporal derivatives. The available context does not establish a final quantitative accuracy, a confirmed maximum-acceleration time, or a verified phoneme association. Those findings must be added only after the corresponding experiments are complete.

11.3 Limitations

Important limitations include potential manual annotation variability, possible ambiguity in contour branches, uncertainty in region closure, the sensitivity of second derivatives to noise, and the need for verified timestamps. A scalar area trajectory cannot capture all spatial aspects of vocal-tract shape. The report also depends on the exact annotation definitions and the approved scope of the underlying dataset.

11.4 Future work

- Complete and document the C1/C2/C3 anatomical annotation schema.
- Improve endpoint selection with topology-aware contour traversal and explicit ambiguity handling.
- Develop unit tests for intersection, polygon construction, and temporal derivatives.
- Compare automated endpoint estimates with expert-reviewed annotations where permitted.
- Quantify area sensitivity to landmark perturbations and alternative region constructions.

- Evaluate suitable smoothing or local-polynomial derivative methods and document peak stability.
- Verify image calibration and report physical units only when supported.
- Establish audio–MRI synchronization from reliable metadata or a validated signal-based procedure.
- Investigate phoneme-level interpretation only after timing validation.
- Package the workflow with configuration files, reproducible outputs, and a concise user guide.

11.5 Final closing statement

The value of the project lies in connecting visual anatomical annotations to transparent, inspectable measurements. A trustworthy result is not simply a number or a peak on a graph; it is a result whose data source, geometry, assumptions, uncertainty, and processing steps can be checked. The final report should reflect both the accomplishments and the unresolved questions of the actual internship work.

ESSENTIAL MATLAB CODE

The snippets below capture core geometric operations for reference.

A. Validate contour data

```
function C = validateContour(C, name)
    if ~isnumeric(C) || size(C,2) ~= 2
        error('%s must be a numeric N-by-2 matrix.', name);
    end
    C = double(C);
    C = C(all(isfinite(C),2), :);
    if size(C,1) < 2
        error('%s has fewer than two valid points.', name);
    end
end
```

B. Remove only consecutive duplicate points

```
function C = removeConsecutiveDuplicates(C)
    if isempty(C) || size(C,1) < 2, return; end
    keep = [true; any(diff(C,1,1) ~= 0, 2)];
    C = C(keep,:);
end
```

This preserves contour traversal order. Do not sort by x or collapse every repeated x coordinate: contours can contain multiple valid y values at the same x.

C. Estimate tangent near a reference landmark

```
dist = vecnorm(C1 - Vm, 2, 2);
[dGap, iStar] = min(dist);
if iStar <= 1 || iStar >= size(C1,1)
    error('Nearest C1 point is at boundary; review tangent.');
end
v = C1(iStar+1,:) - C1(iStar-1,:);
if norm(v) <= eps, error('Degenerate tangent.'); end
t = v / norm(v);
n = [-t(2), t(1)];
if dGap > 3, warning('Reference-to-contour gap is %.3f px.', dGap); end
```

D. Find candidate crossings on C2

```
g = (C2 - Vm) * t(:);
brackets = find(g(1:end-1).*g(2:end) <= 0);
EPs = zeros(0,2);
for m = 1:numel(brackets)
    j = brackets(m);
    den = g(j) - g(j+1);
    if abs(den) <= eps, continue; end
    eta = g(j) / den;
    if eta >= 0 && eta <= 1
        EP = C2(j,:) + eta*(C2(j+1,:) - C2(j,:));
        EPs(end+1,:) = EP; %#ok<AGROW>
    end
end
if isempty(EPs), error('No usable C2 crossing; inspect geometry.'); end
```

```
% If multiple crossings exist, use a documented anatomical rule
% and inspect the selected candidate on the image overlay.
```

E. Area and Acceleration/discrete second difference

```
areaPx2 = polyarea(Xpoly, Ypoly);
A = double(areaPixels(:));
acc = diff(A, 2);
[maxPos, iPos] = max(acc, [], 'omitnan');
[minAcc, iNeg] = min(acc, [], 'omitnan');
framePos = iPos + 1;
frameNeg = iNeg + 1;
```

The indexing above assumes a continuous frame sequence. If failed frames are omitted, preserve original frame IDs explicitly. Units remain pixel²/frame² unless frame timing and spatial calibration are known.

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